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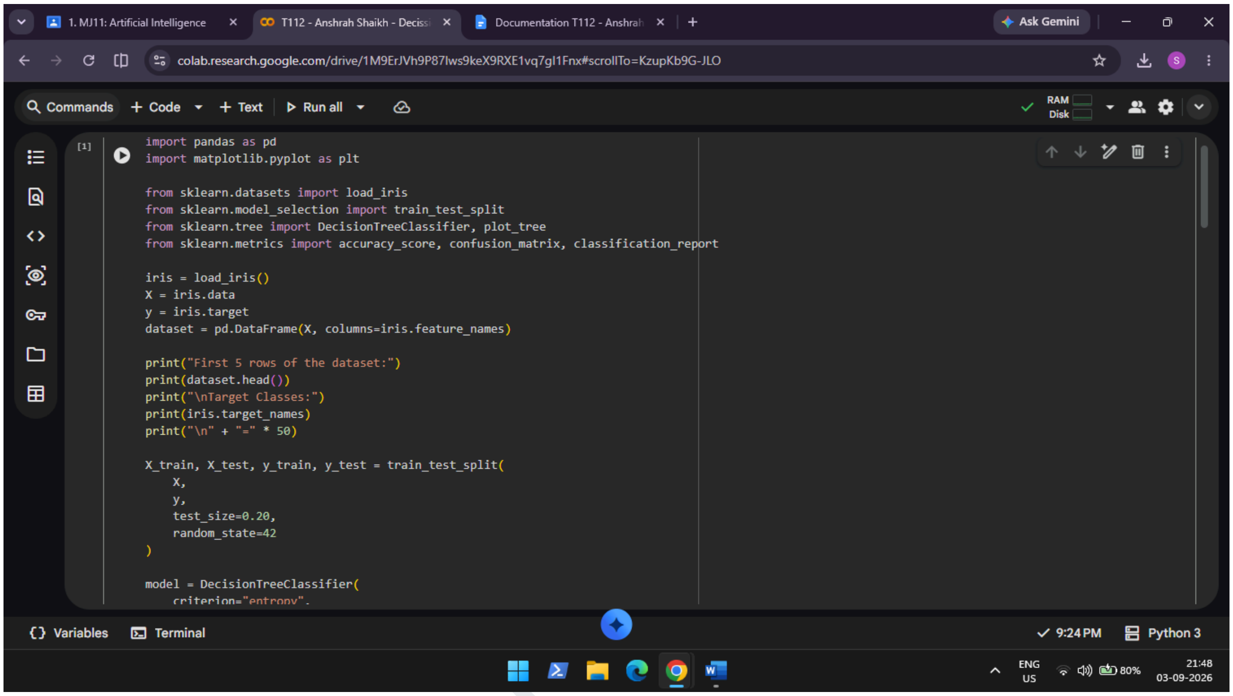
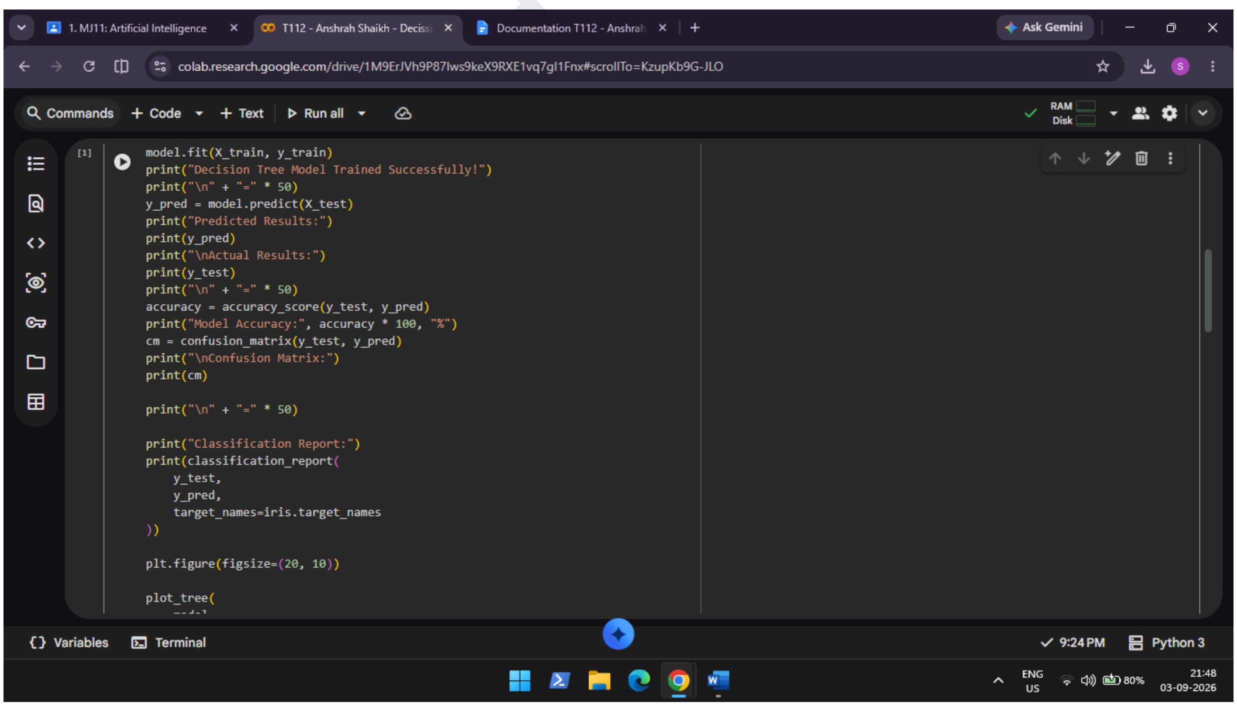
Aim: Decision Tree Learning

- Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
- Evaluate the accuracy and effectiveness of the decision tree on test data.
- Visualize and interpret the generated decision tree.

Input:



```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

iris = load_iris()
X = iris.data
y = iris.target
dataset = pd.DataFrame(X, columns=iris.feature_names)

print("First 5 rows of the dataset:")
print(dataset.head())
print("\nTarget Classes:")
print(iris.target_names)
print("\n" + "=" * 50)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

model = DecisionTreeClassifier(
    criterion="entropy",
```



```python
model.fit(X_train, y_train)
print("Decision Tree Model Trained Successfully!")
print("\n" + "=" * 50)
y_pred = model.predict(X_test)
print("Predicted Results:")
print(y_pred)
print("\nActual Results:")
print(y_test)
print("\n" + "=" * 50)
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", accuracy * 100, "%")
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

print("\n" + "=" * 50)

print("Classification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))

plt.figure(figsize=(20, 10))

plot_tree(
```

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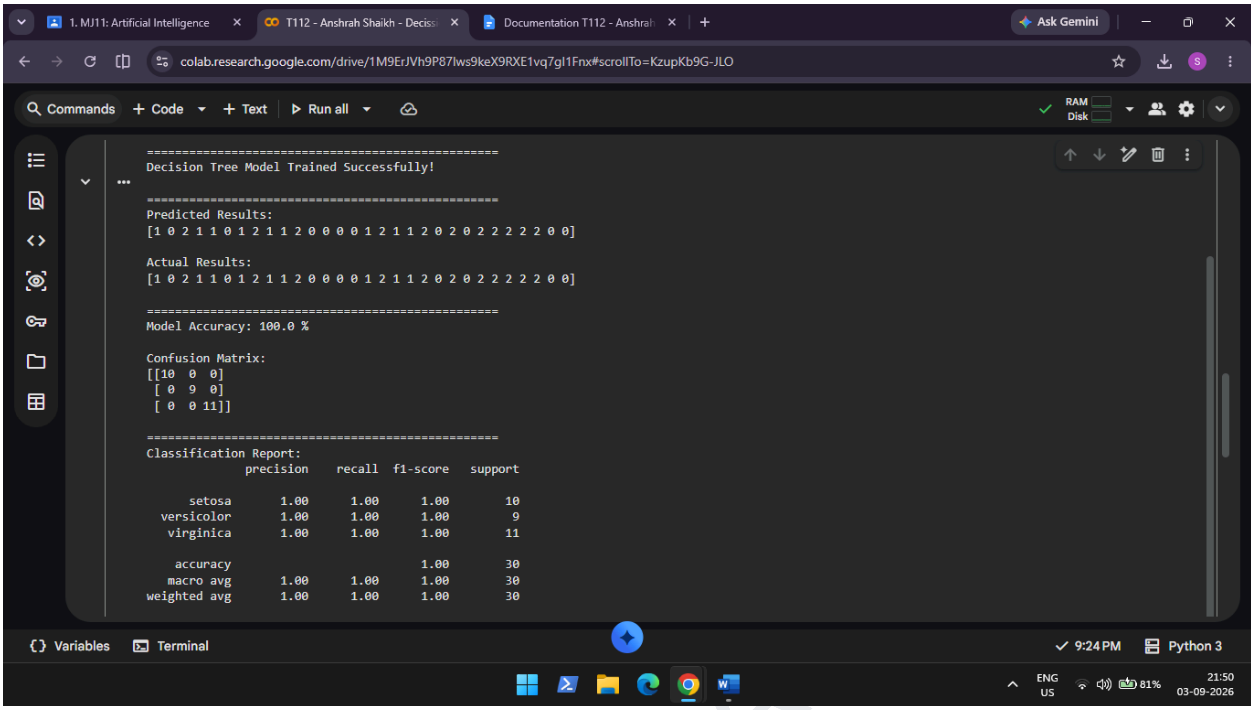
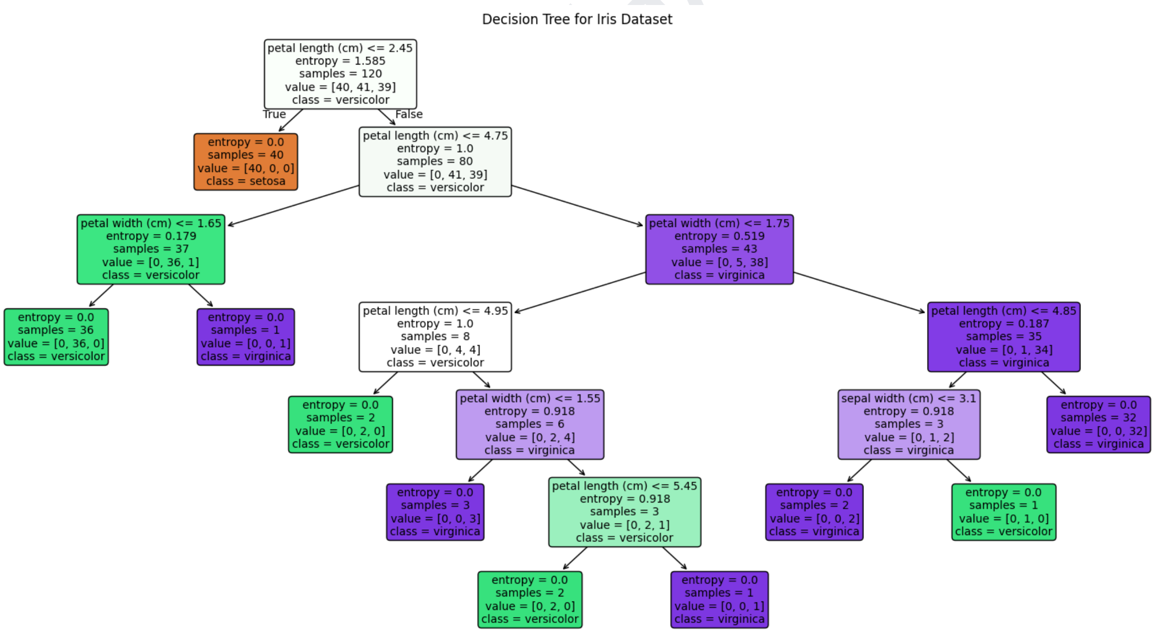
Output:



Visualization:



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